Municipal aggregates recover ninety per cent of the neighbourhood deprivation ordering that satellite imagery supports

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Abstract

Most countries publish social statistics as administrative aggregates and almost none publish them at neighbourhood level. We measure how much neighbourhood detail aggregates and free satellite imagery jointly recover. Training a weakly supervised model on five-level deprivation aggregates of Mexican municipalities — never on any spatial annotation — and validating against fully held-out census-tract grades, the map reaches a within-municipality rank correlation of $\rho = 0.23$, 90% of what the same frozen image features attain under full tract-level supervision ($\rho = 0.25$). Detail rises with every axis of the supervision: more aggregates keep buying map, state-level aggregates halve it, one national figure erases it, and optical content matters where resolution does not. The maps beat the leading downloadable wealth index on identical ground (paired $\Delta\rho = +0.22$), replicate against an independent institution's index ($\rho = 0.36$), close a quarter to two-fifths of the aggregate-to-census gap in simulated programme targeting, and order Bogotá's block-level strata without retraining. Attention-based multiple-instance learning fails at this task for a measurable, general reason.

Introduction

Census-quality social data is expensive, and most of the world sees it roughly once a decade. In between, and below the level the tables are published at, policy runs on aggregates: a municipal index here, a district average there. A rich literature has shown that satellite imagery plus machine learning predicts such indicators across space [1–4], and derived global products already steer programme targeting [3, 5]. But predicting an aggregate somewhere else and *disaggregating it in place* are different tasks, and the second carries the policy weight: a mayor knows their municipality is poor; what they lack is the map of which neighbourhoods are poorest.

We ask the disaggregation question quantitatively: how much neighbourhood detail do a country's published aggregates, combined with free imagery, actually contain? We frame the answer as an exchange rate — so many aggregates, published at such a granularity, buy so much map — and we measure it where measurement is possible. Mexico publishes its Social Deprivation Grade (GRS) both as municipal aggregates and for 61,430 urban census tracts (AGEB) [6]; we train only on the former and score maps against the latter, held out entirely, in cities the model never saw (Fig. 1).

Three design choices keep the measurement honest. Supervision is genuinely weak: a model sees a bag of image tokens for a municipality and one vector — the population shares at each grade threshold, which is what the published aggregate encodes — and no spatial annotation of any kind. Every comparison is scored *within* municipalities, because ordering tracts across cities is mostly solved by knowing which city is poor (pooled correlations run near 0.48 against 0.23 within, at the same operating point). And an oracle trained directly on the held-out tract labels bounds what the frozen representation supports at all, so the quality of weak supervision is always reported as a fraction of a measured ceiling rather than of an imagined perfect map.

Results

The exchange rate

Figure 2 shows the measurement on a pool of 771 municipal bags — the 110 training cities plus 213 municipalities added expressly to cover the deprived half of urban Mexico that large-city samples miss (Fig. 1a; municipalities overlapping any held-out city's ground are excluded, Methods). **Quantity** (Fig. 2a): fifty aggregates buy $\rho = 0.15$ within municipalities; the full national supply buys 0.227 ± 0.010, which is 90% of the 0.251 ceiling the same features reach under full tract supervision. The curve is still rising at the last point: every extra published aggregate keeps buying map. **Granularity** (Fig. 2b): relabelling the same bags with state-level aggregates cuts the map to $\rho = 0.152$; a single national aggregate erases it (0.009). What a statistics office chooses to publish bounds what its users can recover. **Content** (Fig. 2c): block-averaging the optical input to the radar's 20 m effective resolution costs little ($0.183 \rightarrow 0.172$, single-sensor configuration), while radar itself collapses to chance (0.006) — despite having been, in our feature-based phase-one baseline, the best single sensor at detecting high-deprivation tracts (per-city AUROC 0.773 against 0.747 optical). Sensor content, not ground sampling distance, is what fails; the dissociation between predicting and localizing recurs across sensors.

Why the standard tool fails, and what fixes it

The standard weakly supervised localizer — gated-attention multiple-instance learning (AB-MIL [8], with the CLAM instance constraint [9]) — predicts municipal labels well (quadratic-weighted $\kappa = 0.62$) and produces a map at chance (AUROC 0.463; Fig. 3a; these diagnostics ran on the national-set pool). The reason is measurable: flattening the attention to near-uniform with an entropy penalty costs no bag accuracy ($0.899 \rightarrow 0.895$), so the bag task is solvable from the plain feature mean and no gradient ever asks the attention to point anywhere.

Inverting the composition fixes it. Predicting every token and averaging the *predictions* — deep learning from label proportions [10–12] — makes the map the mechanism rather than a by-product: AUROC 0.761 under the same features, bags and labels, rising to 0.802 after a hyperparameter sweep and 480 m of neighbourhood context. Because Mexican municipal grades are population-weighted aggregates of tract grades, the setting is label proportions with *known* instance weights; weighting each token by its tract's census population is a robustness variant that performs equivalently ($\rho = 0.232$ against 0.227).

The sweep exposed a structural blindness worth naming. Across fourteen configurations, the rank correlation between bag-level error — the only model-selection signal weak supervision may legitimately use — and map quality is -0.14 (Fig. 3b): honest model selection is nearly blind to the quantity that matters. Whether this extends beyond settings where the bag label is an average of instance states is an open question, but any pipeline whose product is the map — computational pathology included — has to confront it, and it is why our benchmark ships with held-out instance labels.

External validity

Four independent checks argue the map measures deprivation rather than an artifact (Fig. 4). Against the leading downloadable product, Meta's Relative Wealth Index [3], scored per tract on an identical universe of 3,191 AGEB with a city-clustered bootstrap of the paired difference, the map wins in 18 of 22 municipalities: $\Delta\rho = +0.22$ $[+0.08, +0.29]$ (Fig. 4a). A land-cover-plus-NDVI head with no learned representation reaches $\rho = 0.149$, so the foundation-model features carry signal beyond built-up density. Against an institution it never trained toward — CONAPO's continuous urban marginalization index, built from a different indicator set over the same census — the map correlates at $\rho = 0.361 \pm 0.073$ within municipalities, higher than against its own five-level training construct, as a tie-free target should allow (Fig. 4b). In simulated geographic targeting with fractional funding of the marginal tract, allocating by the map instead of the municipal aggregate closes 26–43% of the gap to census-guided allocation at budgets of 10% of population and above, and *loses* to the aggregate at the tightest budget (5%), where errors at the very top of the ranking are unforgiving (Fig. 4c). The map is also not an administrative artifact: crossing a municipal border where deprivation does not change moves it by $+0.019$ $[-0.002, +0.155]$, and tract borders inside municipalities are crossed smoothly ($+0.001$); an earlier $+0.14$ border jump traced to bag-limited neighbourhood adjacency — our inference seam, corrected, not the model's shortcut.

The construct travels; a different construct does not

Applied unchanged to Bogotá, the model orders Colombia's block-level socioeconomic strata at $\rho = 0.318$ (AUROC 0.716 for the deprived strata 1–2 over 14,132 tokens joined to blocks; Fig. 5). Applied to Rio de Janeiro, it detects IBGE's mapped informal settlements barely above chance (AUROC 0.557 among built-up tokens). The contrast is informative rather than disappointing: Colombian strata measure the same construct as the GRS — a graded deprivation ordering over the whole urban fabric — while the AGSN mask delimits informality, a partly orthogonal notion. Weak supervision on aggregates taught a deprivation gradient, and the gradient, not a country signature, is what transfers.

Limits

The ceiling is the honest one: even with full tract labels and neighbourhood context, 160 m tokens support $\rho = 0.251$ within municipalities — useful for ordering neighbourhoods, far from reading individual blocks. Agreement depends strongly on tract size: $\rho = 0.18$ for tracts covered by 1–3 tokens, rising to 0.45 at 20–49 tokens, and falling back to 0.10 in the two municipalities dominated by very large tracts — the modifiable areal unit problem stated in units an office can act on. Ecological caution applies throughout: the model recovers where deprived tracts are, and tract grades are themselves aggregates over households; neither the labels nor the maps identify deprived households, and the targeting simulation counts people in deprived tracts, not deprived people.

Discussion

The exchange rate reframes a familiar technology. Rather than another demonstration that imagery predicts poverty, it answers the question a statistics office would ask: *what does the resolution of publication buy at the resolution of inference?* The measured axes give the answer operational shape — the national supply of municipal aggregates recovers 90% of what these features support, coarser publication forfeits map in direct proportion, optical content rather than resolution is the binding sensor property — and the targeting experiment prices the result in people reached, crossover included. The negative results carry equal weight: attention-based MIL fails here for a structural reason; aggregate-level validation is nearly blind to map quality; and informality detection is not a free by-product of a deprivation gradient. Each is a warning any weakly supervised mapping effort will meet.

Mexico's dual-level publication made the calibration possible; the measured fractions are what travel to countries that publish only aggregates. Bogotá is the first step of that argument. Extending the dual calibration to Colombia's own strata — fine truth held out, coarse aggregates built from it — is the natural next measurement, and the released benchmark makes every number here re-runnable from frozen features in minutes.

Methods

Imagery. Sentinel-2 L2A and Sentinel-1 RTC annual median composites for 2020 (10 m grid, windowed COG reads from Microsoft Planetary Computer), for 424 boxes covering the 138 cities of the national set and the 213-municipality expansion; city boxes are conurbations and typically span several municipalities. **Labels.** CONEVAL GRS 2020: municipal-level population shares at each of four grade thresholds supervise training; AGEB-level grades are used only for evaluation and for the oracle. Auxiliary evaluation sources: CONAPO IMU 2020 [7], ESA WorldCover 2020 [14], Meta RWI [3], IBGE AGSN 2019 [15], Bogotá's district stratification [16]. **Bags and instances.** A bag is a municipality (771 after exclusions); instances are 16-pixel tokens (160 m) encoded by a frozen DOFA foundation model [13], both sensors fused per token. Municipalities contributing ground to any held-out city are excluded from the expansion — an earlier version leaked nine validation municipalities this way, detectable only at bag level, and every affected number was re-measured. **Model.** A two-layer head on standardized frozen features predicts four cumulative-grade probabilities per token; the bag prediction is the mean of token predictions, fit with binary cross-entropy; each token's hidden representation is concatenated with the mean of its 8 grid neighbours (480 m), computed city-wide at inference. **Protocol.** 110/14/14 cities for train/validation/test, stratified by size and deprivation, no city split across sets; the test cities remain unopened in this draft and will be scored once, with every reported row, before submission. Sweeps select on grouped training folds under bag error only; validation cities are scored once per reported configuration; comparisons are paired by municipality over three seeds; every interval on a within-municipality mean is a city-clustered bootstrap (2,000 resamples), and curve error bars are s.d. over seeds. **Oracle.** The same head trained on tract labels, with the same context, bounds the representation ($\rho = 0.251$ with context, 0.202 without). **Zero-shot transfer.** Bogotá and Rio get the same 2020 composite protocol; tokens are scored by the three final seeds with Mexican standardization statistics, deliberately unadapted. Rio's evaluation is restricted to tokens with WorldCover built fraction ≥ 0.10. **Data and code.** The full pipeline, frozen vectors, labels, splits and protocol are released; every figure regenerates from one canonical results file.

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Data availability

The frozen benchmark — foundation-model vectors for every bag, municipal and tract labels, splits and the evaluation protocol — will be deposited on Zenodo with a DOI at publication. All source data are public: Copernicus Sentinel via Microsoft Planetary Computer, CONEVAL, CONAPO, INEGI, ESA WorldCover, Meta RWI (HDX), IBGE, and Bogotá's open-data portal.

Code availability

The full pipeline (MIT licence) is available at the project repository, pinned by `uv.lock`; each figure maps to one driver script.

Competing interests

The author declares no competing interests.

Figures

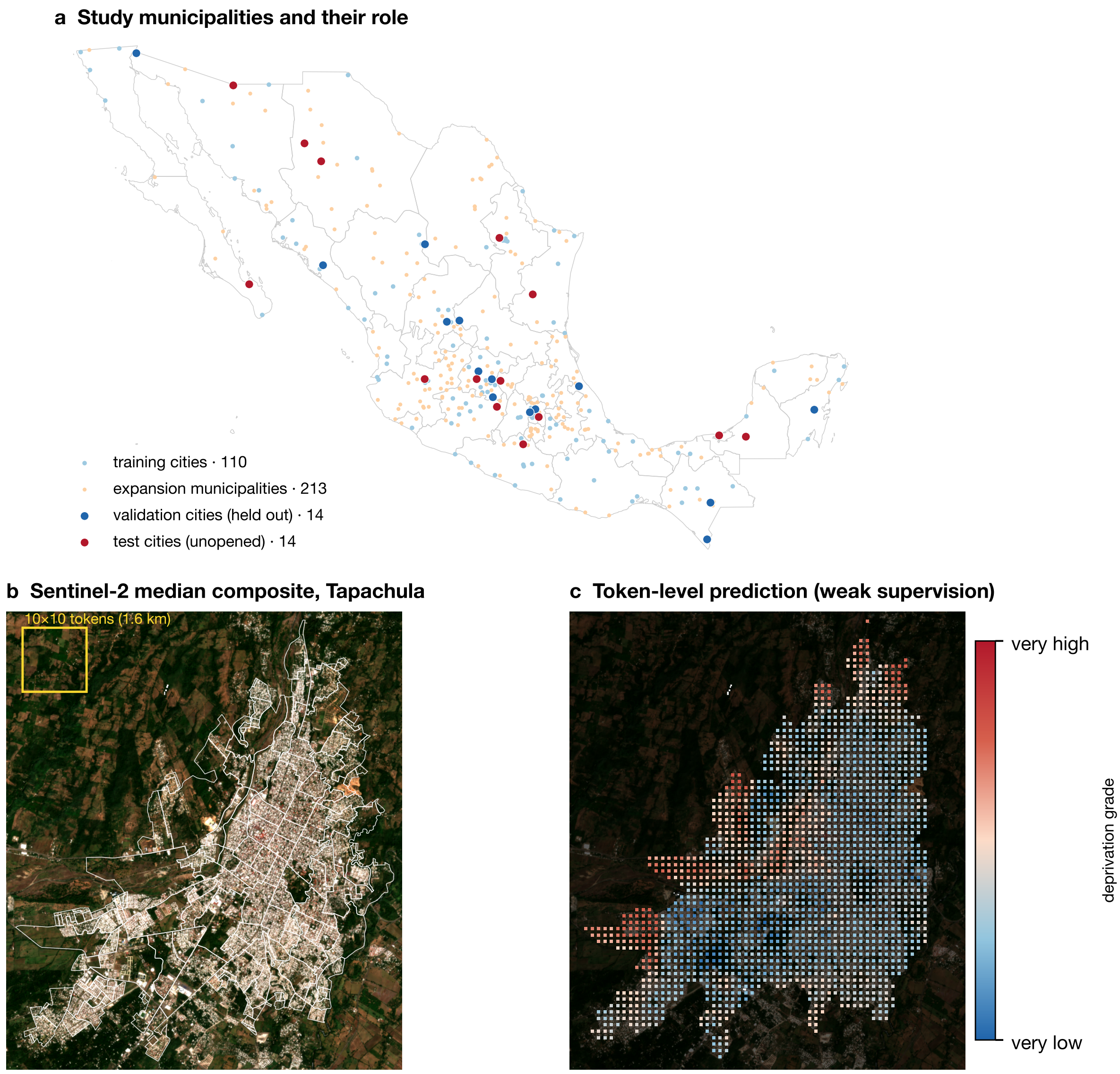

Figure 1: **Study design.** **a**, The 351 study municipalities: 110 training cities, the 213-municipality expansion toward the deprived half of urban Mexico (municipalities overlapping held-out ground excluded), 14 validation cities and 14 unopened test cities. **b**, What the model consumes: the 2020 Sentinel-2 median composite of Tapachula (a validation city) with AGEB boundaries overlaid; tokens of 160 m tile the box. **c**, What it produces: the token-level deprivation score from municipal aggregates alone. Tract ground truth for the same city is shown in Supplementary Fig. 1; Tapachula sits at the validation median ($\rho = 0.47$), chosen so the example neither flatters nor sandbags the method.

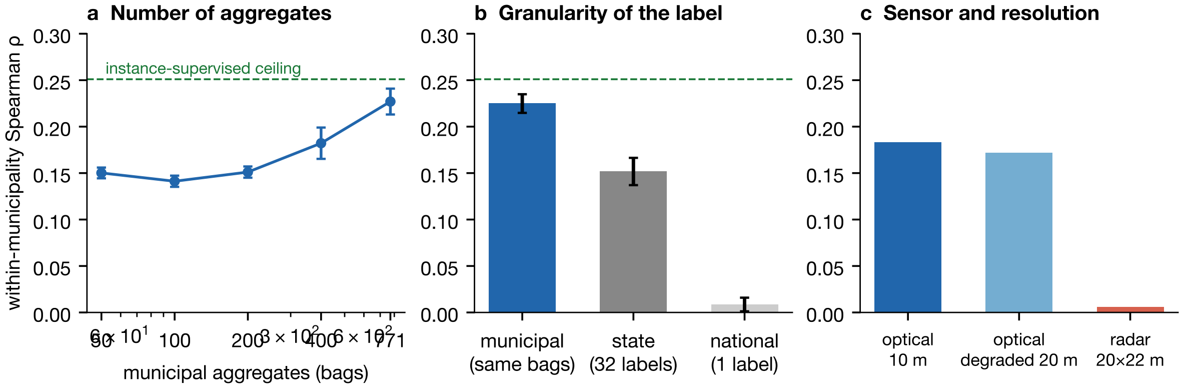

Figure 2: **The exchange rate between aggregates and maps.** All panels: mean over three seeds on the 14 validation cities, against held-out tract grades; dashed line, the instance-supervised oracle on the same features and context ($\rho = 0.251$). **a**, Agreement as a function of how many municipal aggregates supervise training (nested, grade-stratified sampling; error bars, s.d. over seeds). **b**, The same 771 bags relabelled with coarser aggregates (error bars, s.d. over seeds). **c**, Single-sensor configurations: destroying optical resolution to the radar's grain barely hurts; radar collapses.

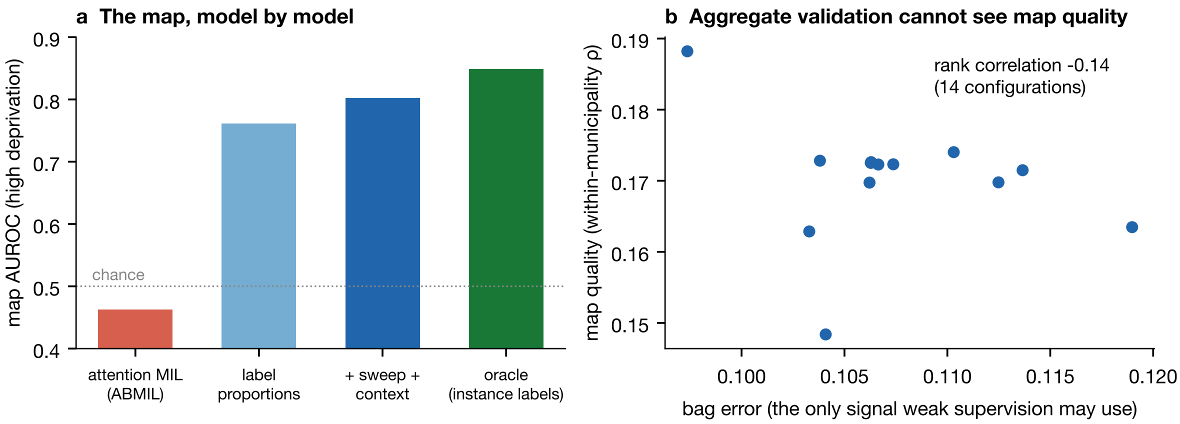

Figure 3: **Predicting is not localizing. a**, Map quality by model family on identical features and labels (national-set pool for the ABMIL diagnostics). Attention pooling solves the bag task from the feature mean and receives no localization gradient; predicting instances and averaging predictions makes the map the mechanism. **b**, Across the hyperparameter sweep, the only criterion weak supervision may monitor is nearly uncorrelated with map quality.

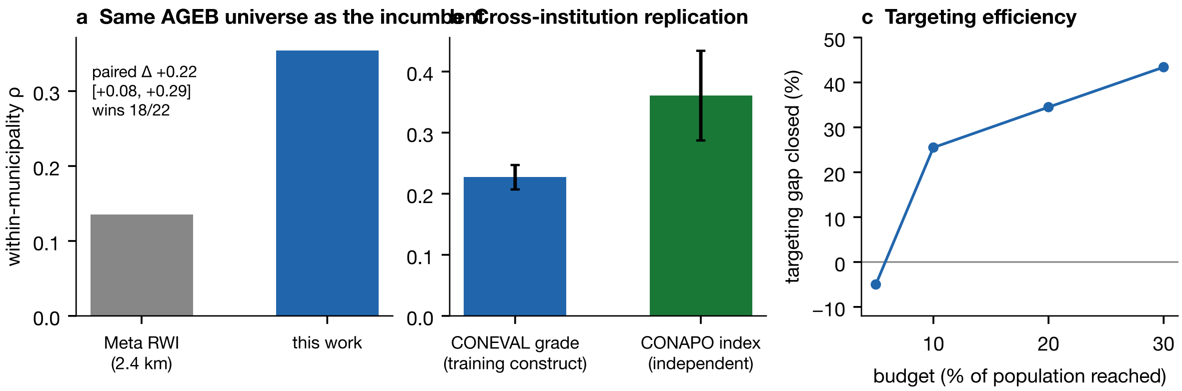

Figure 4: **External validity.** **a**, Within-municipality agreement on the identical AGEB universe reached by the incumbent's grid; annotation, city-clustered bootstrap of the paired difference. **b**, Replication against an independent institution's continuous index (error bars, city-clustered bootstrap half-widths). **c**, Share of the aggregate-to-census targeting gap closed at each budget, people pooled across the 14 validation cities; the map loses at the tightest budget and wins from 10% upward.

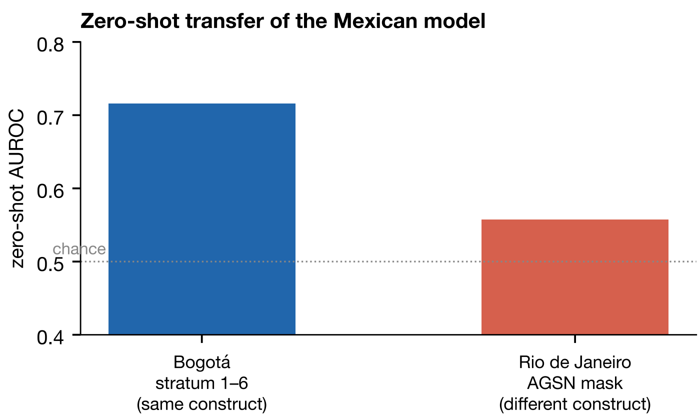

Figure 5: **Zero-shot transfer.** The Mexican model, unchanged and unadapted, ordering Bogotá's block strata (AUROC for strata 1–2; n = 14,132 tokens) and detecting Rio's mapped informal settlements among built-up tokens (n = 34,331). The deprivation gradient travels; informality is a different construct.